Jeromy Chang

San Francisco, Bay Area | (415) 823-3381 | jchang5@scu.edu | linkedin.com/in/jeromy-chang

Education

- **Santa Clara University** — Santa Clara, CA
  Master of Science in Artificial Intelligence (MSAI), GPA: 3.914 — *Jun 2026*
  - **Selected Graduate Coursework:** NLP, Deep Learning; ML/Data Mining & Pattern Recognition; Optimization; Advanced Algorithms; Image/Video Compression; Advanced Topics - Visual AI
- **Santa Clara University** — Santa Clara, CA
  Bachelor of Science in Computer Science & Engineering, GPA: 3.622 — *Jun 2025*
  - **Honors:** SCU Academic Scholarship (2021–Present); Hack for Humanity 2022 – Best Use of Blockchain

Research Experience

- **Multilingual Deception Detection (In Progress)** — Winter 2026 – Present
  LLM Generation & Human-Likeness Evaluation (Industry-mentored; SCU + Amazon)
  - Built a reproducible pipeline for generating **multilingual** hotel reviews across **10 languages** using controlled settings (0-shot vs few-shot, Non-English vs English prompts, varying LLM generators) with standardized run tracking.
  - Developed **LLM-as-judge** evaluation (single + pairwise) with **reliability controls** (**A/B swaps**, **position-bias checks**, **bootstrap CIs**) and tracked detector performance (**ROC AUC/F1**).
  - Implemented **experiment logging/scoreboards** (standardized run IDs, configs) to enable reliable comparison across models/prompts; tracked **duplication** and **distribution deltas** as quality gates.
- **GRADE: Generalizable Reasoning-Aware Dialogue Evaluation for AI Tutors** — Spring 2026
  Conference paper submitted to EMNLP 2026; under review - NLP/LLM Evaluation
  - Extended the BEA 2025 TutorMind benchmark from mistake identification to four pedagogical dimensions plus joint multitask prediction; evaluated 120 configurations across five **open-source LLMs**, **zero-shot/LoRA**, **reasoning modes**, and **generated/self-verified augmentation**.
  - Reached 0.77 strict macro-F1 on mistake identification, outperforming the strongest prior BEA 2025 system by **5 macro-F1 points**; identified model-dependent augmentation effects and a failure mode in which LoRA interfered with structured instruction following under thinking mode.
- **LoRA-MIC: Low-Rank Adaptation for Machine-Vision Image Compression** — Winter 2026
  *Paper accepted to **IEEE MMSP 2026***
  - Proposed LoRA-MIC, injecting low-rank adapters into conv + attention layers of a frozen transformer-based image codec (TIC), training only **∼0.25M params** (∼3.3% of TIC).
  - Achieved **50.76% BD-Rate savings** (detection) and **52.33%** (segmentation) on **COCO 2017**, outperforming TransTIC with **∼6× fewer** trainable parameters.
- **Santa Clara University School of Engineering** — Santa Clara, CA
  Senior Design Research – SparseInfer (Optimizing LLM Inference) — *Sep 2024 – Jun 2025*
  - Collaborated in a team of three to expand on SparseInfer research paper and **enhance inference speed** of LLMs (Llama2) on edge devices by **10%**, leveraging **activation sparsity** without accuracy degradation.
  - Engineered a dynamic CPU-GPU communication channel employing a PID controller for real-time hyperparameter fine-tuning, significantly improving adaptability and inference speed.

Industry Experience

- **Nutanix** — San Jose, CA
  AI/ML Engineering Intern (SaaS Engineering) — *May 2025 – Aug 2025*
  - Deployed to production a lifecycle-based "case aging" feature for an in-house escalation prediction model, improving **F1 by ∼5 points** (to ∼25%) and exceeding the previous baseline.
  - Ranked **10 candidate signals** using EDA, correlation analysis, **SHAP explainability**, and precision/recall/F1 to prioritize high-signal features.
  - Built a **REST API** exposing live/training model metrics for self-serve monitoring; resolved data reliability issues (e.g., stale/idle-time logic) across Salesforce + **Postgres/MongoDB**.
  - Hackathon: Built a **RAG-assisted** unit test generation service (API + prompting) producing runnable Python tests and returning pass/fail + coverage.

Skills

Technical: Python, C/C++, SQL, Java, PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SHAP, Postgres, MongoDB
Languages: English, Cantonese